



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Worksheet 5

1. Given the potential:

$$V = \frac{10}{r^2} \sin \theta \cos \phi$$

- Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.
 - Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point A($1, 30^\circ, 120^\circ$) to B($4, 90^\circ, 60^\circ$).
2. Two dipoles with dipole moments $-5\mathbf{a}_z \text{ nC/m}$ and $9\mathbf{a}_z \text{ nC/m}$ are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.
3. Three point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, respectively. Find the energy in the system.